

COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM(S) TO BE EVALUATED **BS-CS FALL 2023**

Course Description

Course Code	CS-1002	
Course Title	Programming Fundamentals	
Credit Hours	3+1	
Prerequisites by Course(s) and Topics	N/A	
Grading Policy	Absolute grading	
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.	
Policy about late submission of assessment item	Penalties for late submissions of assignment/project: - Up to 30 mins late, loss of 40% of the mark awarded. - After 30 mins, assignment/project will not be accepted for marking.	
Course Plagiarism Policy	Plagiarism in project or midterm or final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignment category for both students.	
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assessment items of Theory Part	
	Assessment Item	Number
	Assignments	3
	Sessional - I	1
	Sessional - II	1
	Quizzes	>= 5
	Project	1
	Final Exam	1
Course Instructors	Dr. Arshad Islam (Section A, B), Mr. Shehreyar Rashid (Section E, F, G), Mr. Aqib Rehman (Section C, D)	
Lab Instructors (if any)		
Course Coordinator	Dr. Arshad Islam	
URL (if any)	Class Link: https://classroom.google.com/c/NjE4NTQ5ODc2MTc1?cjc=yt2dhnq	



	Class code: yt2dhng
Current Catalog Description	The course aims to equip students with the basic computing concepts and to provide them the ability to analyze the given requirements for solving problems in different domains while implementing the solutions on a computer system. It emphasizes on developing an algorithm and applying the basic programming constructs like control structures, arrays, functions, pointers, dynamic memory allocation, etc. for its development. The students will learn the syntax of the C++ programming language for the implementation.
Textbook (or Laboratory Manual for Laboratory Courses)	Tony Gaddis "STARTING OUT WITH C++" 9 th Edition
Reference Material	Paul Deitel, Harvey Deitel "C++ How to Program" 10 th Edition Walter Savitch "Problem Solving with C++" 10 th Edition

Course Learning Outcomes	A. Course Learning Outcomes (CLOs)		
	After completion of the course, the students shall be able to: 1. Understand basic problem-solving steps and logic constructs. 2. Apply basic programming concepts. 3. Design and implement algorithms to solve real-world problems.		
	B. Program Learning Outcomes		
	For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.		
	1. Computing Knowledge:	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems	✓
	2. Problem Analysis:	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓
	3 Design/Develop Solutions:	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓
	4. Investigation & Experimentation:	Conduct investigation of complex computing problems using research based knowledge and research based methods.	
	5. Modern Tool Usage:	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	
	6. Society Responsibility:	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.	
	7. Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems.	
	8. Ethics	Apply ethical principles and commit to professional ethics and responsibilities and	

		norms of computing practice.												
		9. Individual and Teamwork:	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.										✓	
		10. Communication	Communicate effectively on complex computing activities with the computing community and with society at large											
		11. Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member of a team.											
		12. Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.											
C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)														
		PLOs												
			1	2	3	4	5	6	7	8	9	10	11	12
CLOs	1	✓	✓											
	2		✓	✓										
	3			✓							✓			

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	List of Topics	No. of Weeks	Contact Hours
	- Problem-solving, Basic flowchart, block diagram, and programming languages. - Primitive data types, input/output (hello world). - Signed and unsigned data types, constants and variables.	1	3
	- Arithmetic operators (+, -, *, /, % and their compound counterparts) with their associativity and precedence. - Bit wise operators	2	6
	- Conditional/selection structures. - Comparison and logical operators. - if, if. . else and if else if structure. - Switch statement, <i>break</i> statement. - Ternary operator.	2	6
	1st Sessional Examination		
	- Repetition structures. - Pre/post increment/decrement operators. - while loop (sentinels + condition). - Loop with <i>for</i> . - Loop with <i>do-while</i> . - Nesting of <i>while</i> , <i>for</i> loop and <i>continue</i> statement.	3	9
	- Function prototypes, definition, and calling.	1	3
	- Introduction to Arrays. - Array initialization and representation. - Char arrays. - Multi-Dimensional Arrays (MDA). - MDA representation in memory.	2.33	7
	2nd Sessional Examination		
	- Aliases, parameters passing by value and by reference (passing arrays). - Function calling order and stack (function within a function). - Recursion	1.66	5
	- Pointers. - const. vs. non-const. pointers, a pointer to const. data vs. a pointer to non-constant data. - Using pointers. - Dynamic memory allocation. - Array of pointers.	2	6
	Total	15	45
Laboratory	Yes, there are lab tasks with every lab of three hours.		



Projects/Experiments Done in the Course				
Programming Assignments Done in the Course	Yes, there are six programming assignments and a project.			
Class Time Spent (in hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	34	5	5	1
Oral and Written Communications	Every student is required to submit at least __1__ written reports of typically __5__ pages and to make __1__ demonstration of typically __10__ minutes duration.			